**National Institute of Technology Silchar, Assam**

**Department of Electronics and Communication Engineering.**

Project Title :

# Diabetic Retinopathy detection using Machine Learning Algorithms.

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# Literature Reviews

| Title of the Research Paper | Author(s) | Key findings |
|---|---|---|
| A deep learning based model for diabetic retinopathy grading | Akhtar S et al | The research achieved the development of a high-performance deep learning model called RSG-Net that automates the detection and severity grading of the disease |
| Machine Learning-based Diabetic Retinopathy Early Detection and Classification Systems | Dathar A. Hasan et al | The study finds that machine learning, particularly the ResNet50 deep convolutional neural network, effectively detects and classifies Diabetic Retinopathy (DR) severity. |

# Literature Reviews

| Title of the Research Paper | Author(s) | Key findings |
|---|---|---|
| Diabetic Retinopathy: Present and Past | Gupta A et al | The presence and combination of specific lesions—such as red dots (microaneurysms), shiny white deposits (hard exudates), and yellow/white superficial lesions (cotton wool spots)—are used to determine the severity of the disease. |
| Diagnosis of diabetic retinopathy using machine learning techniques | R Priya et al | Among the models tested, the Support Vector Machine (SVM) consistently outperformed others, achieving the highest diagnostic accuracy of 97.6%. |
| | | |

# Literature Reviews

| Title of the Research Paper | Author(s) | Key findings |
|---|---|---|
| Diabetic Retinopathy Detection using Machine Learning | R Revathy et al | The study proposes an automated hybrid machine learning model that extracts features like exudates and haemorrhages to detect diabetic retinopathy with 82% accuracy. By combining classifiers such as SVM, KNN, and Random Forest, the system achieved a precision of 0.8119 and effectively identified 36 out of 49 test samples correctly |
| DREAM: Diabetic Retinopathy Analysis Using Machine Learning | Roychowdhury S et al | Researchers found that the Gaussian Mixture Model (GMM) is most effective for classifying bright lesions, while the k-Nearest Neighbor (kNN) algorithm performs best for red lesions. |

# Background Study

Diabetic Retinopathy (DR) is a diabetes-related eye disease that damages the blood vessels of the retina and is one of the leading causes of blindness among the working-age population worldwide. According to the World Health Organization, approximately 347 million people worldwide suffer from diabetes, and nearly 93 million are affected by diabetic retinopathy. The Centers for Disease Control and Prevention estimates that 29.1 million people in the United States have diabetes, with nearly 40–45% developing some stage of DR.

DR progresses through five stages:

- 0 – No DR
- 1 – Mild
- 2 – Moderate
- 3 – Severe
- 4 – Proliferative DR

Early detection is critical because vision impairment can be slowed or prevented if treatment is initiated at the right time. However, DR often shows no early symptoms, making timely diagnosis difficult.

# Background Study

Currently, DR detection relies on manual examination of color fundus retinal images by trained ophthalmologists. This process is:

- Time-consuming
- Expensive
- Dependent on specialist availability
- Prone to delays and human error

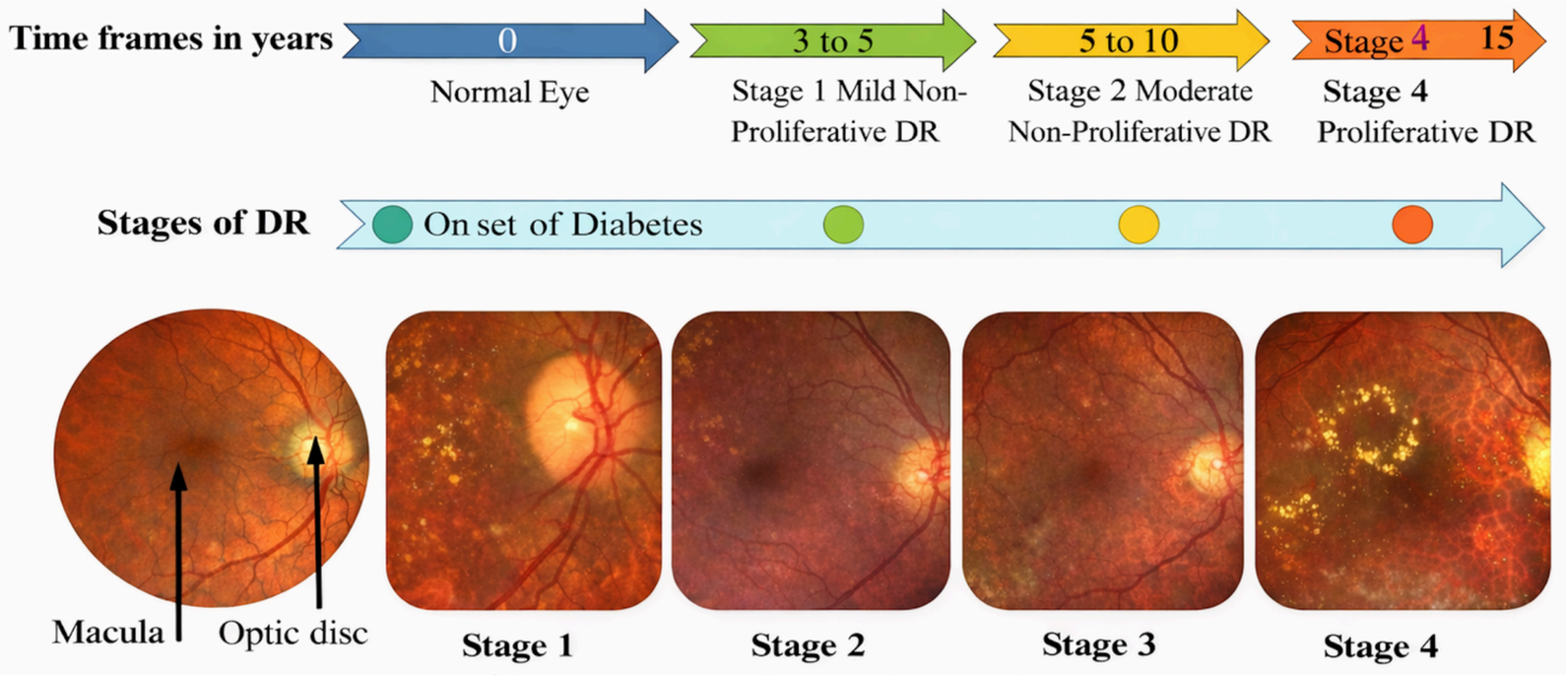


Figure 1. Stages of diabetic retinopathy.

# Problem Statement

Diabetic Retinopathy detection currently depends on manual evaluation of retinal fundus images by expert clinicians, which is time-consuming, resource-intensive, and not scalable in high-population or rural areas.

The major challenges include:

- Shortage of trained ophthalmologists
- High screening cost
- Delayed diagnosis leading to late treatment
- Large volume of retinal images requiring evaluation
- Class imbalance in DR datasets

There is a strong need to develop an automated, accurate, and scalable machine learning-based system that can classify retinal images into different DR severity stages with clinical reliability.

# Objectives

Dataset Collection: To obtain a large dataset of retinal fundus images (e.g., EyePACS dataset) containing labelled DR severity stages.

Image Preprocessing

- Image resizing and normalisation
- Noise reduction
- Contrast enhancement
- Data augmentation

Model Development

To design and train multiple machine learning and deep learning models, such as:

- Transfer Learning model ResNet50

# Objectives

Performance Evaluation

To evaluate models using:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

Model Optimization

To improve generalisation and reduce overfitting using:

- Data augmentation
- Hyperparameter tuning

# Major Findings

- Deep learning models, particularly CNN-based architectures, significantly outperform traditional machine learning algorithms in DR image classification tasks.
- Transfer learning using pretrained models (e.g., ResNet50) improves accuracy due to feature extraction from large-scale image datasets.
- Data preprocessing and augmentation greatly enhance model robustness.
- Quadratic Weighted Kappa is a more reliable metric than accuracy alone for medical grading problems.
- Automated systems can achieve performance comparable to human graders in detecting moderate to severe DR.

# Research Gaps

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Despite promising advancements, several gaps remain:

1. **Imbalanced Dataset Problem**
   Most DR datasets have a high number of “No DR” images compared to severe stages, leading to biased model predictions.
2. **Lack of Generalization**
   Models trained on a specific dataset (e.g., EyePACS) may not generalize well to other populations or imaging devices.
3. **Explainability Issue**
   Many deep learning models act as black boxes, lacking interpretability required for clinical adoption.
4. **Real-Time Deployment Challenges**
   Most research focuses on offline evaluation rather than real-time hospital integration.
5. **Limited Focus on Mild Stage Detection**
   Early-stage DR detection remains difficult due to subtle retinal lesions.

# Methodology

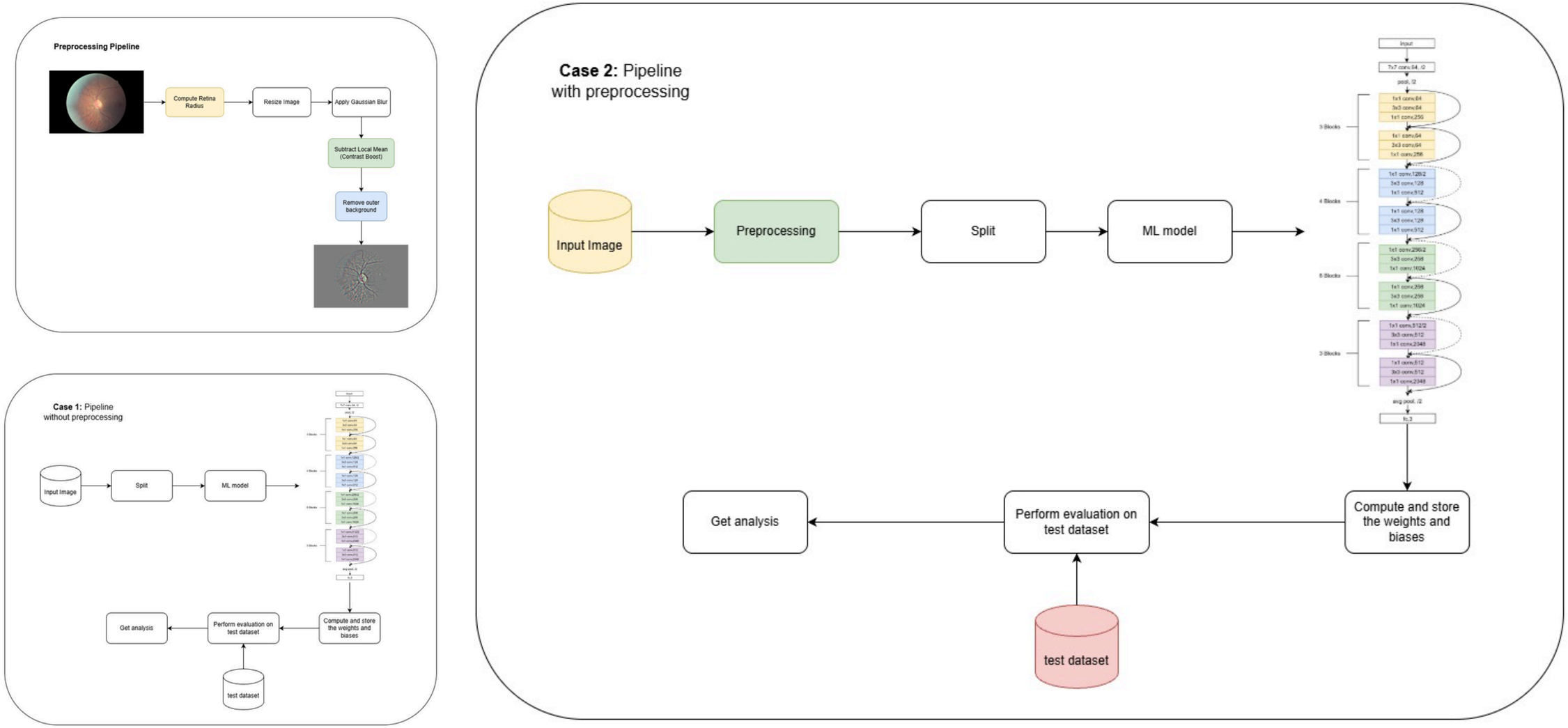


Fig 2: Complete Methodology flow diagram.

# Work Done

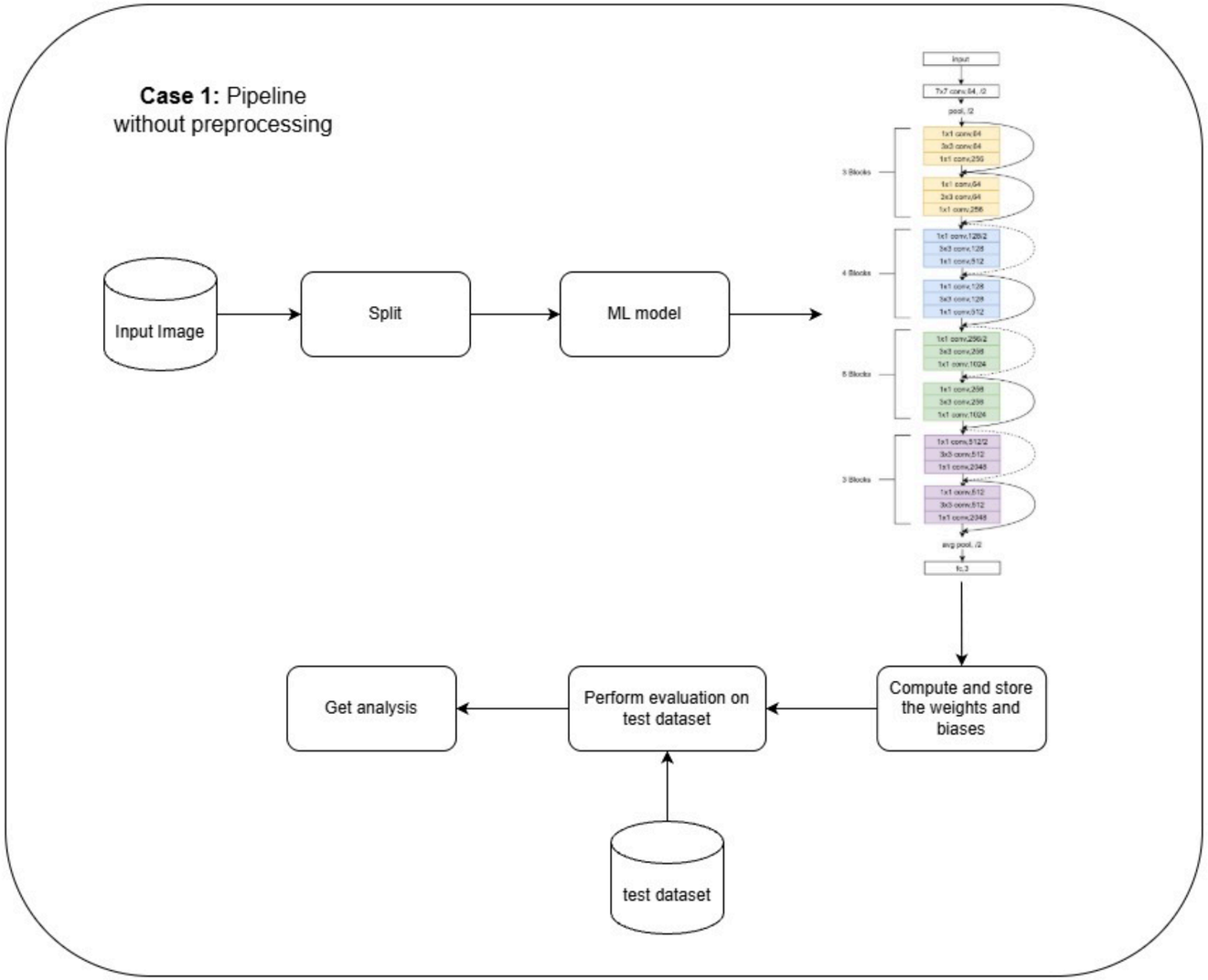


Fig 3: Pipeline without preprocessing

# Work Done

```
Overall Test Accuracy: 76.654776060245 74% (1934/2523)
Quadratic Weighted Kappa: 0.5449457877588282

Classification Report:
              precision    recall  f1-score   support

           0       0.82      0.95      0.88      1845
           1       0.17      0.08      0.11       177
           2       0.56      0.38      0.45       385
           3       0.58      0.33      0.42        66
           4       0.62      0.16      0.25        50

    accuracy                           0.77      2523
   macro avg       0.55      0.38      0.42      2523
weighted avg       0.72      0.77      0.73      2523
```

Fig 4: Pipeline without preprocessing metrices

# Work Done

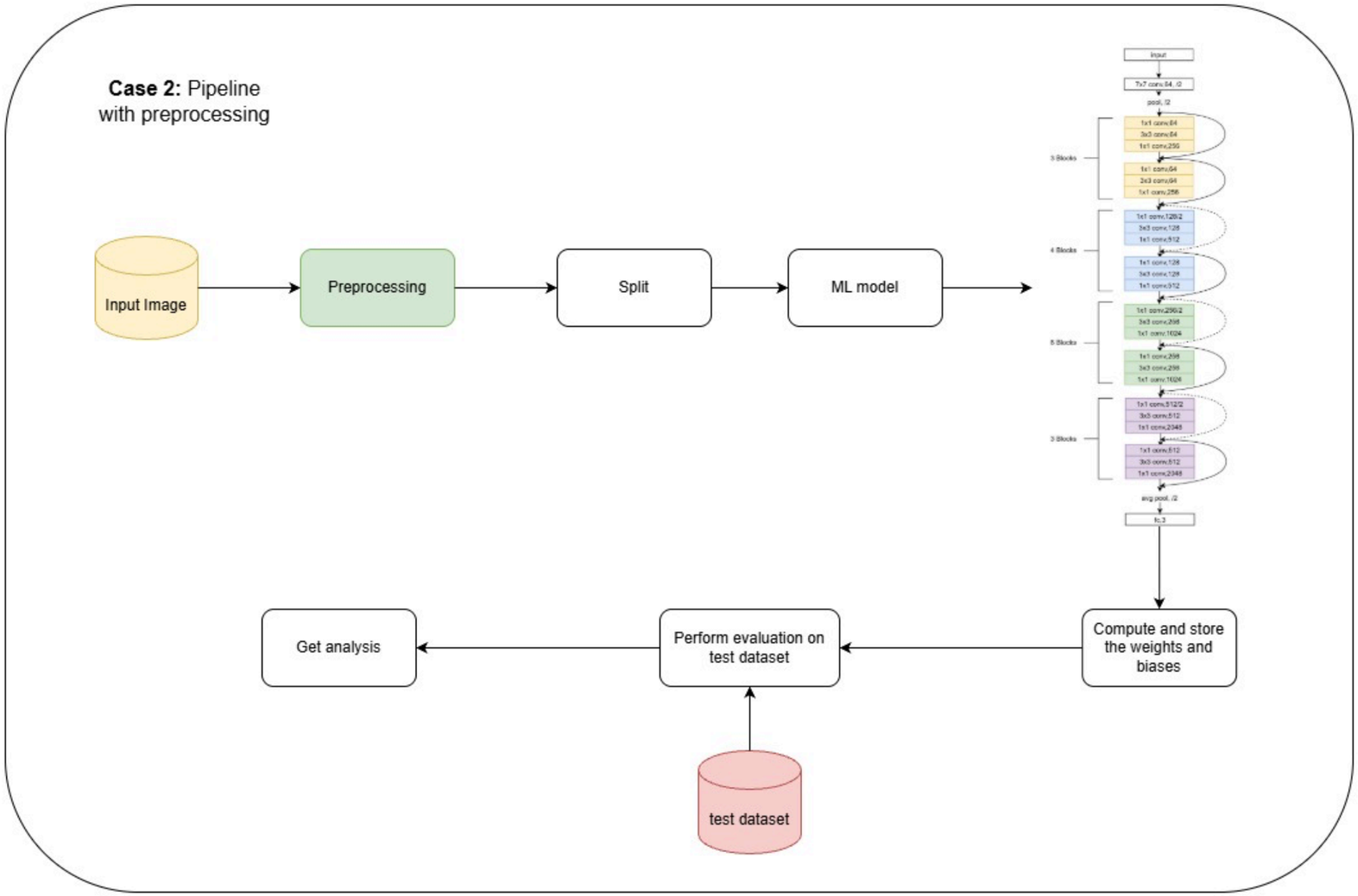


Fig 5: Pipeline with preprocessing

# Work Done

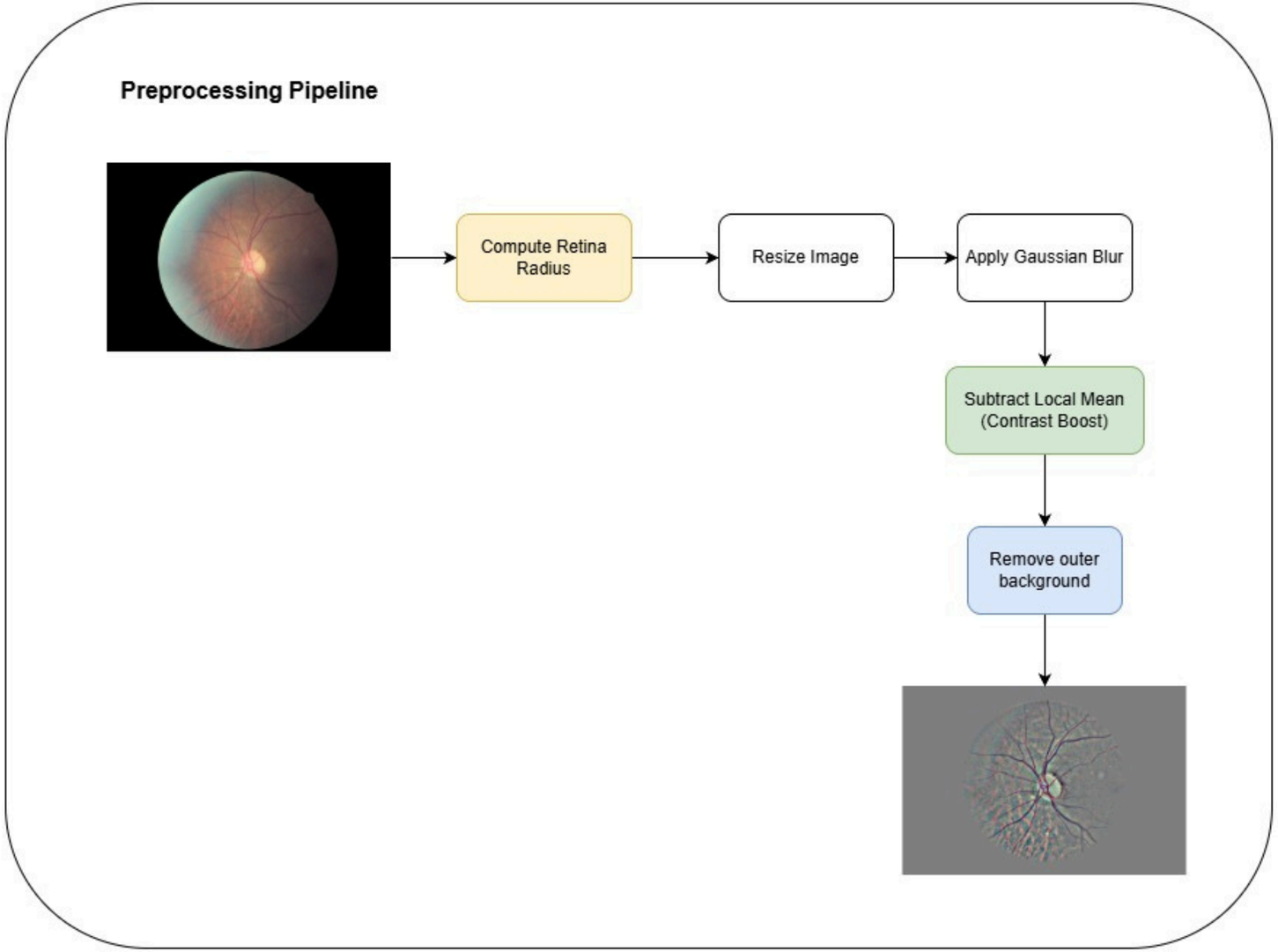


Fig 6: Image preprocessing pipeline

# Work Done

- Collected and prepared a labeled retinal fundus image dataset for five-class Diabetic Retinopathy severity classification (0–4).
- Performed image preprocessing including resizing, normalization, and data augmentation.
- Trained a deep learning model for multi-class DR detection.
- Evaluated performance using Accuracy, Precision, Recall, F1-score, Confusion Matrix, and Quadratic Weighted Kappa (QWK).
- Achieved **87.03% test accuracy** and **0.74 QWK score**.
- Analyzed confusion matrix and identified performance limitations in Severe (Class 3) and Proliferative (Class 4) DR stages.

# Result and Analysis

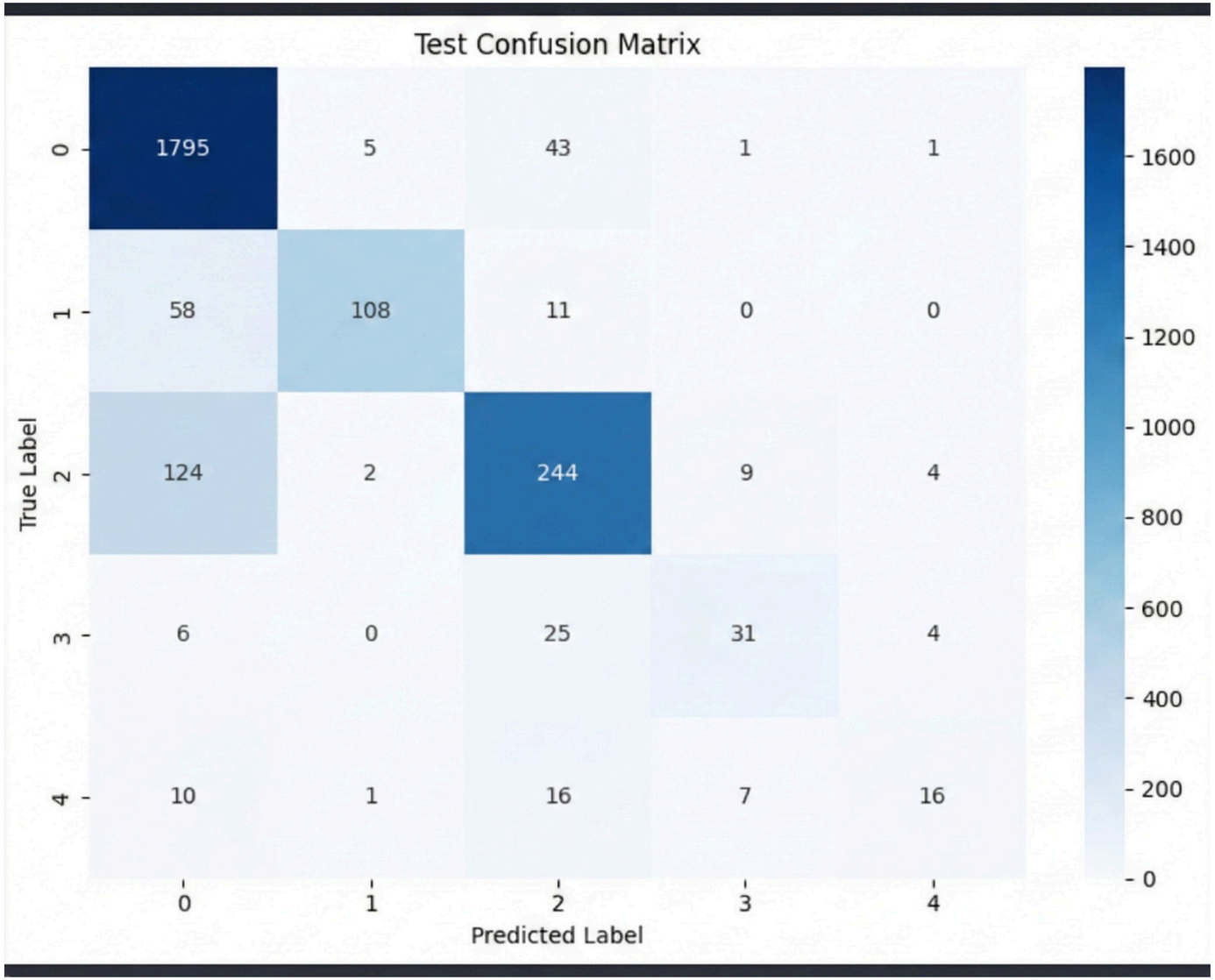


Fig 7: Performance Metrics Comparison for Multi Classification.

# Result and Analysis

```
Overall Test Accuracy: 87.02896% (2194/2521)
Quadratic Weighted Kappa: 0.740398321012345

Classification Report:
              precision    recall  f1-score   support

           0       0.90      0.97      0.94      1845
           1       0.93      0.61      0.74       177
           2       0.72      0.64      0.68       383
           3       0.65      0.47      0.54        66
           4       0.64      0.32      0.43        50

    accuracy                           0.87      2521
   macro avg       0.77      0.60      0.67      2521
weighted avg       0.86      0.87      0.87      2521

    accuracy       0.87                0.87      2521
   macro avg       0.77      0.60      0.67      2521
weighted avg       0.86      0.87      0.87      2521
```

Fig 8: Performance Metrics for Multi-class Classification.

# Result and Analysis

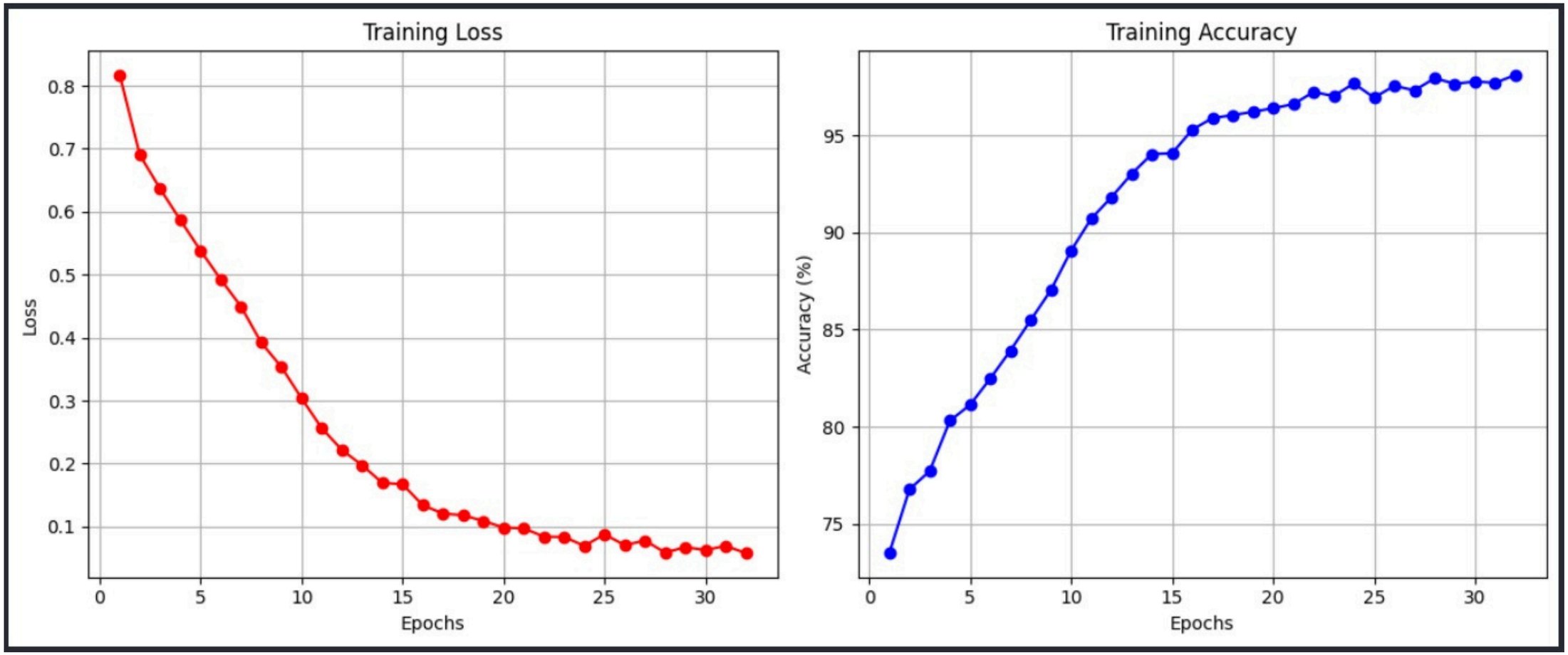


Fig 9: Epoch-wise Training Loss and Accuracy Progression

# Outcome

Based on the experimental results obtained from the trained Diabetic Retinopathy (DR) detection model, the following outcomes were achieved:

1. Overall Model Performance

- Test Accuracy: 87.03% (2194 correct predictions out of 2521 images)
- Quadratic Weighted Kappa (QWK): 0.74

The Quadratic Weighted Kappa score of 0.74 indicates substantial agreement between the model predictions and human expert grading. Since QWK is the primary evaluation metric for DR grading, this confirms that the model performs reliably for clinical-style severity classification.

2. Confusion Matrix Insights

From the confusion matrix:

- Most predictions are concentrated along the diagonal, indicating correct classification.
- The largest confusion occurs:
    - Between Class 2 and Class 0
    - Between Class 3 and Class 2
    - Between Class 4 and Class 2

# Outcome

3. Training Performance Analysis

From the training curves:

- Training Loss steadily decreases from ~0.82 to ~0.06, indicating effective learning.
- Training Accuracy increases from ~73% to ~98%, showing strong convergence.
- No major instability or oscillation is observed.
- The gap between training and test accuracy (~98% vs 87%) suggests mild overfitting but still acceptable generalisation.

# Conclusion:

Diabetic Retinopathy is a major global health concern requiring early and accurate diagnosis to prevent blindness. Manual screening methods are insufficient to handle the growing diabetic population.

This project demonstrates that machine learning and deep learning algorithms can effectively analyze retinal fundus images and classify DR severity with high accuracy and reliability. The use of advanced CNN architectures and transfer learning significantly enhances detection performance.

Automated DR detection systems have strong potential to:

- Reduce diagnostic delays
- Minimize human workload
- Improve accessibility in rural areas
- Prevent vision loss through early intervention

With further improvements in interpretability and deployment strategies, machine learning-based DR screening systems can become a vital component of future ophthalmic healthcare infrastructure.

# Future Work

- The confusion matrix shows that the model did not achieve acceptable performance for Class 3 (Severe DR) and Class 4 (Proliferative DR), mainly due to misclassification and class imbalance. In future work, we will focus on improving detection accuracy for these advanced stages using better class balancing and model optimization techniques.

- Additionally, we plan to implement **CLAHE (Contrast Limited Adaptive Histogram Equalization)** for enhanced image preprocessing to improve retinal feature visibility and achieve better classification results.

# References


**[1]** Akhtar S. et al , proposed a high-performance deep learning model named RSG-Net, which automates both detection and severity grading of diabetic retinopathy. Their model demonstrated strong classification capability, improving reliability in clinical diagnosis.

**[2]** Dathar A. Hasan et al , conducted a comprehensive survey on machine learning-based DR detection systems and concluded that deep convolutional neural networks, particularly ResNet50, show superior performance in detecting and classifying DR severity.

**[3]** Revathy R. et al, developed a hybrid machine learning model that extracts pathological features such as exudates and haemorrhages. By combining classifiers including SVM, KNN, and Random Forest, their system achieved 82% accuracy with a precision score of 0.8119.

**[4]** Roychowdhury S. et al. introduced the DREAM framework for DR analysis. Their findings showed that the Gaussian Mixture Model (GMM) effectively classifies bright lesions, while the k-Nearest Neighbor (kNN) algorithm performs best in detecting red lesions.

# References


**[5]** Gupta A. et al. analyzed the clinical aspects of diabetic retinopathy and highlighted that lesion types such as microaneurysms, hard exudates, and cotton wool spots are key indicators for determining disease severity.

**[6]** Priya R. et al. evaluated multiple machine learning models and found that the Support Vector Machine (SVM) achieved the highest diagnostic accuracy of 97.6%, outperforming other classifiers.

**[7]** Emma Dugas, Jared, Jorge, and Will Cukierski. Diabetic Retinopathy Detection. https://kaggle.com/competitions/diabetic-retinopathy-detection, 2015. Kaggle.

**[8]** EyePACS, “EyePACS: Free platform for retinopathy screening.” [Online]. Available: https://www.eyepacs.com.

# THANK YOU